

Enterprise Identity & Access Management using Microsoft Entra ID and Azure RBAC

Final Project Report

Executive Summary

This project implements an enterprise-grade Identity and Access Management (IAM) solution on Microsoft Azure following Zero Trust and least-privilege principles. It demonstrates secure identity management, RBAC, Managed Identity, Azure Key Vault, Storage Security, Monitoring, and Automation.

Project Objectives

- Implement Microsoft Entra ID for centralized identity management.
- Control access using Azure RBAC and custom roles.
- Secure secrets with Azure Key Vault and Managed Identity.
- Protect Azure Storage using RBAC, HTTPS and TLS 1.2.
- Monitor resources using Azure Monitor and Log Analytics.
- Automate administration using Azure CLI and Azure PowerShell.

Implemented Components

- Microsoft Entra ID users and groups
- Azure RBAC role assignments
- Custom RBAC roles
- Resource Groups
- Virtual Network and Network Security Groups
- Windows Virtual Machine
- System-assigned Managed Identity
- Azure Key Vault
- Azure Storage Account and Blob Containers
- Azure Monitor and Log Analytics Workspace
- Azure CLI automation
- Azure PowerShell automation

Security Controls

- Zero Trust Architecture
- Least Privilege Access
- Group-based Authorization
- Passwordless Authentication with Managed Identity
- Private Blob Containers
- HTTPS Only and TLS 1.2
- Security Monitoring and Activity Logging

- RBAC Validation and Security Testing

Testing & Validation

- Validated RBAC permissions for HR, Finance, Development and Security groups.
- Verified Managed Identity access to Key Vault.
- Verified private Storage access using Azure RBAC.
- Verified NSG rules restricting RDP access.
- Verified Activity Logs and Log Analytics data collection.

Automation

- Azure CLI scripts for resource management.
- Azure PowerShell scripts for inventory, RBAC reporting and security audits.

Risk Assessment

- Credential theft mitigated using Managed Identity.
- Unauthorized access mitigated using Azure RBAC.
- Secret exposure mitigated using Azure Key Vault.
- Network exposure reduced using NSGs.
- Monitoring enabled for auditing and incident investigation.

Skills Demonstrated

- Microsoft Entra ID
- Azure RBAC
- Azure Key Vault
- Managed Identity
- Azure Storage Security
- Azure Monitor
- Log Analytics
- KQL
- Azure CLI
- Azure PowerShell
- Zero Trust
- Cloud Security

Conclusion

- The project successfully demonstrates an enterprise-ready Azure IAM implementation aligned with cloud security best practices. It showcases identity governance, access control, monitoring, automation, and documentation suitable for an entry-level Cloud Security Engineer or Identity & Access Administrator portfolio.